MODULE 6.3

Configure a ZPF

Zone-Based Policy Firewalls | CLI Configuration, Step by Step

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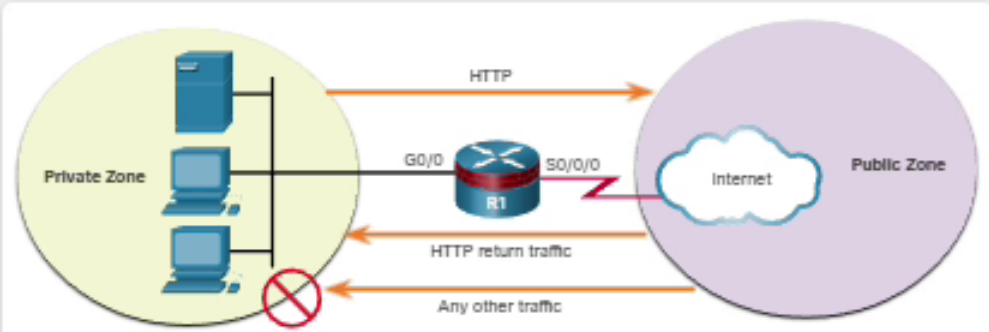
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Configure a ZPF: Overview

- **Five steps, not always in order:** create the zones, identify traffic with a class-map, define an action with a policy-map, identify a zone-pair and match it to a policy, then assign zones to the interfaces.
- **Some steps do depend on order:** a class-map must exist before it can be assigned to a policy-map, and a policy-map must exist before it can be assigned to a zone-pair — skipping ahead gets you a router error message.
- **One topology throughout:** the same R1 topology is used for every example in this module — HTTP (plus HTTPS/DNS) is allowed from the Private zone to the Public zone, matching return traffic is allowed back, and everything else from the Public zone is blocked.



Step 1: Create the zones.
Step 2: Identify traffic with a class-map.
Step 3: Define an action with a policy-map.
Step 4: Identify a zone pair and match it to a policy-map.
Step 5: Assign zones to the appropriate interfaces.

Step 1: Create the Zones

- **Before creating zones, answer three questions:** what interfaces should be included, what will each zone be named, and what traffic is necessary between the zones and in which direction.
- **This topology:** two interfaces, two zones, traffic flowing in one direction — traffic sourced from the public zone will not be allowed.
- **Command:** create each zone on the firewall with the zone security command.

```
Syntax:
Router(config)# zone security zone-name
```

```
Example – R1:
R1(config)# zone security PRIVATE
R1(config-sec-zone)# exit
R1(config)# zone security PUBLIC
R1(config-sec-zone)# exit
R1(config)#
```

Step 2: Identify Traffic — Syntax

- **A class:** a way of identifying a set of packets based on its contents, using "match" conditions — defined with class-maps.
- **For ZPF, use the inspect keyword.** match-any means packets need to meet just one match criterion; match-all means they must meet every one.
- **Match traffic** to an ACL, a specific protocol, or even another class-map.

Syntax – create the class-map:

```
Router(config)# class-map type inspect [match-any | match-all] class-map-name
```

Syntax – match statements:

```
Router(config-cmap)# match access-group {acl-# | acl-name}
Router(config-cmap)# match protocol protocol-name
Router(config-cmap)# match class-map class-map-name
```

match access-group: matches by ACL. match protocol: matches by named protocol. match class-map: nests another class-map's criteria.

Step 2: Identify Traffic — Example

- **This topology:** HTTP traffic is being allowed to cross R1 from the PRIVATE to the PUBLIC zone.
- **Best practice:** when allowing HTTP traffic, it's recommended to specifically include HTTPS and DNS too, since web browsing depends on both.
- **match-any:** traffic can match any one of the three statements below to become a member of the HTTP-TRAFFIC class.

Example – R1:

```
R1(config)# class-map type inspect match-any HTTP-TRAFFIC
R1(config-cmap)# match protocol http
R1(config-cmap)# match protocol https
R1(config-cmap)# match protocol dns
```

Step 3: Define an Action — Syntax

A policy-map defines what action to take for traffic that's a member of a class. An action is specific functionality tied to a traffic class — inspect, drop, and pass are the three actions.

```
Syntax:
R1(config)# policy-map type inspect policy-map-name
R1(config-pmap)# class type inspect class-map-name
R1(config-pmap-c)# {inspect | drop | pass}
```

Action	What it does
inspect	State-based traffic control — the router maintains session info for TCP and UDP, and permits return traffic.
drop	Discards unwanted traffic. This is the default action for all traffic.
pass	A stateless action that allows the router to forward traffic from one zone to another.

Step 3: Define an Action — Example

- **This example:** the HTTP-TRAFFIC class from Step 2 is associated with a new policy-map named PRIV-TO-PUB-POLICY, with the inspect action applied.
- **inspect:** if traffic traveling PRIVATE → PUBLIC is inspected, R1 maintains session state and automatically permits the matching return traffic.
- **drop (implicit):** every policy-map ends with an implicit class class-default that drops everything that isn't a member of an explicit class — unlike ACLs, this is silent, with no ICMP unreachable sent back.
- **pass:** ideal for secure, predictable protocols like IPsec — but it's one-directional, so a return policy must be configured separately. Most application traffic is better handled with inspect.

Example – R1:

```
R1(config)# policy-map type inspect PRIV-TO-PUB-POLICY
R1(config-pmap)# class type inspect HTTP-TRAFFIC
R1(config-pmap-c)# inspect
```

Step 4: Zone-Pair and Match to a Policy

- **Identify a zone pair:** a zone-pair names a source zone and a destination zone, and its own self keyword specifies the router itself (going to or from the router).
- **Then attach the policy:** use service-policy type inspect to bind the policy-map and its action to that zone-pair.
- **This example:** a zone-pair named PRIV-PUB is created with PRIVATE as the source and PUBLIC as the destination, then the policy-map from Step 3 is attached to it.

Syntax:

```
Router(config)# zone-pair security zone-pair-name source {source-zone-name | self}
    destination {destination-zone-name | self}
Router(config-sec-zone-pair)# service-policy type inspect policy-map-name
```

Example – R1:

```
R1(config)# zone-pair security PRIV-PUB source PRIVATE destination PUBLIC
R1(config-sec-zone-pair)# service-policy type inspect PRIV-TO-PUB-POLICY
```

Step 5: Assign Zones to Interfaces

- **The last step:** assign each zone to its interface — this immediately applies whatever service-policy is associated with that zone.
- **Careful:** if no service-policy is yet configured for a zone, all transit traffic through that interface will be dropped.
- **Command:** use zone-member security in interface configuration mode.
- **This example:** GigabitEthernet 0/0 is assigned to PRIVATE, and Serial 0/0/0 is assigned to PUBLIC. Once both are assigned, the service-policy is active: HTTP/HTTPS/DNS traffic from PRIVATE to PUBLIC is inspected, and PUBLIC-to-PRIVATE traffic is only allowed if it's part of a session PRIVATE already started.

```
Syntax:
Router(config-if)# zone-member security zone-name
```

```
Example – R1:
R1(config)# interface GigabitEthernet 0/0
R1(config-if)# zone-member security PRIVATE
R1(config-if)# interface Serial 0/0/0
R1(config-if)# zone-member security PUBLIC
```

Verify a ZPF Configuration: show run

Verify by viewing the running configuration. The class-map comes first, then the policy-map uses it. Notice the highlighted class class-default, which drops all traffic that isn't a member of HTTP-TRAFFIC.

```
R1# show run | begin class-map
!
class-map type inspect match-any HTTP-TRAFFIC
  match protocol http
  match protocol https
  match protocol dns
!
policy-map type inspect PRIV-TO-PUB-POLICY
  class type inspect HTTP-TRAFFIC
   inspect
  class class-default
   drop
!
zone security PRIVATE
zone security PUBLIC
zone-pair security PRIV-PUB source PRIVATE destination PUBLIC
 service-policy type inspect PRIV-TO-PUB-POLICY
!
interface GigabitEthernet0/0
 zone-member security PRIVATE
!
interface Serial0/0/0
 zone-member security PUBLIC
```

Verify a ZPF Configuration: Live Session

After a PRIVATE zone host (192.168.1.3) establishes an HTTPS session with a web server (10.1.1.2), this command shows the match counters and the live, stateful session — plus four packets that hit class-default and were dropped.

```
R1# show policy-map type inspect zone-pair sessions

policy exists on zp PRIV-PUB
  Zone-pair: PRIV-PUB
  Service-policy inspect : PRIV-TO-PUB-POLICY
  Class-map: HTTP-TRAFFIC (match-any)
    Match: protocol http    12 packets, 384 bytes
    Match: protocol https   5 packets, 160 bytes
    Match: protocol dns     0 packets, 0 bytes
  Inspect
    Number of Established Sessions = 1
    Session 2204E220 (192.168.1.3:1049)=>(10.1.1.2:443) https:tcp
    SIS_OPEN/TCP_CLOSEWAIT
      Created 00:00:14, Last heard 00:00:11
      Bytes sent (initiator:responder) [821:1431]
  Class-map: class-default (match-any)
    Match: any
    Drop   4 packets, 160 bytes
```

Verify a ZPF Configuration: Targeted Commands

Four more commands give a view of specific portions of the configuration:

```
show class-map type inspect
Class Map type inspect match-any
  HTTP-TRAFFIC (id 1)
    Match protocol http
    Match protocol https
    Match protocol dns
```

```
show zone security
zone self
  Description: System Defined Zone
zone PRIVATE
  Member Interfaces: GigabitEthernet0/0
zone PUBLIC
  Member Interfaces: Serial0/0/0
```

```
show zone-pair security
Zone-pair name PRIV-PUB
  Source-Zone PRIVATE Destination-Zone PUBLIC
  service-policy PRIV-TO-PUB-POLICY
```

```
show policy-map type inspect
Policy Map type inspect PRIV-TO-PUB-POLICY
  Class HTTP-TRAFFIC
    Inspect
  Class class-default
    Drop
```

ZPF Configuration Considerations

- The router never filters traffic between interfaces in the same zone.
- An interface cannot belong to multiple zones. To union security zones, specify a new zone with its own policy map and zone pairs.
- ZPF can coexist with Classic Firewall, although they can't be used on the same interface — remove the ip inspect interface command before applying zone-member security.
- Traffic can never flow between an interface assigned to a zone and one without a zone assignment. Applying zone-member always causes a temporary service interruption until the other zone-member is configured.
- The default inter-zone policy is to drop all traffic, unless specifically allowed by the zone-pair's service-policy.
- The zone-member command does not protect the router itself — traffic to and from the router is unaffected unless the zone-pair is configured using the predefined self zone.

Summary

- **Five steps:** create zones (zone security) → identify traffic (class-map type inspect) → define an action (policy-map type inspect) → bind a zone-pair to that policy (zone-pair security + service-policy) → assign zones to interfaces (zone-member security).
- **Order matters partway through:** a class-map must exist before it's used in a policy-map, and a policy-map must exist before it's attached to a zone-pair.
- **Three actions:** inspect (stateful, tracks sessions), drop (the implicit class-default on every policy-map), and pass (stateless, one-directional).
- **Verification:** show run confirms the static configuration; show policy-map type inspect zone-pair sessions shows it actually working, live, against real traffic.
- **6.3.9 —** zone-member security is the step that activates everything — and it's also the step most likely to cause an outage if the policy isn't fully configured first.

References

Cisco Networking Academy — Network Defense, Module 6.3: Configure a ZPF

https://www.netacad.com

Cisco — Zone-Based Policy Firewall Design and Application Guide

https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/98628-zone-design-guide.html

Cisco — Zone-Based Policy Firewall Command Reference

https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/sec_data_zbf/command/sec-zbf-cr-book.html